1 Cauchy–Euler equations
- Announcements
- Cauchy–Euler equations
- Worked examples
- Practice problems



Mathematics
and Statistics

$$\int_M d\omega = \int_{\partial M} \omega$$

Mathematics 2C03
Introduction to Differential Equations

Instructor: David Earn

Lecture 22
Cauchy–Euler equations
Tuesday 3 November 2026

Announcements

- The TA is Illya Kierkosz (`kierkosi@mcmaster.ca`)
  - Office hours by appointment: e-mail to arrange a meeting.
- Course web site: `https://davidearn.github.io/math2c/`
- The course outline is on the Simple Syllabus web site.
- Avenue to Learn provides secure access to Echo360 live streams and lecture recordings.
- Assignments and class polls are on `childsmath`.
- **e-mail:** PLEASE always include 2C03 in the subject line of any message about this course, and PLEASE indicate what the issue is. For example:
  - `Subject: Math 2C03: confusion about assignment 1`
  - `Subject: Math 2C03: conflict with test window`
- Office hour: Thursdays, 9:30–10:20 am, after class.

Cauchy–Euler Equations

[Trench §7.4, starting at Definition 7.4.2].

[Trench Theorem 7.4.3 and its proof].

Note: Trench [Definition 7.4.2] calls the homogeneous second-order case an *Euler equation*. We also use *Cauchy–Euler* and *equidimensional*.

Cauchy-Euler Equations

What other types of linear ODEs can we solve?

- We have so far found ways to solve linear ODEs with *constant coefficients*.
- In general, with *variable coefficients*, we cannot express solutions using familiar functions.
- But some variable coefficient linear ODEs that frequently arise can be solved, similar to the way we solved constant coefficient equations.

Cauchy-Euler Equations

Consider the nth order ODE

$$L[y] = a_n x^n \frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_{n-1} x^{n-1} \frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \cdots + a_1 x \frac{\mathrm{d}y}{\mathrm{d}x} + a_0 y = g(x)$$

where $a_n, a_{n-1}, \ldots, a_1, a_0 \in \mathbb{R}$ are constants with $a_n \neq 0$, and quantities in red are functions of the independent variable x.

ODEs that can be written in the above form are known as ***Cauchy-Euler equations*** or ***Equidimensional equations***.

Note:

- In a Cauchy-Euler equation, the coefficient of the nth derivative of y is a constant (a_n) times x^n.
- The coefficient of $\frac{\mathrm{d}^n y}{\mathrm{d}x^n}$ is zero for $x = 0$, so our basic existence-uniqueness theorem for linear ODEs [Trench Theorem 9.1.1] does not apply on intervals that include $x = 0$.

Cauchy-Euler Equations

$$L[y] = a_n x^n \frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_{n-1} x^{n-1} \frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \cdots + a_1 x \frac{\mathrm{d}y}{\mathrm{d}x} + a_0 y = g(x) \tag{CE}$$

where $a_n, a_{n-1}, \ldots, a_1, a_0 \in \mathbb{R}$ are constants with $a_n \neq 0$, and quantities in red are functions of the independent variable x.

We begin by solving the associated homogeneous equation, $L[y] = 0$, on the interval $\mathcal{I} = (0, \infty)$ to avoid $x = 0$.

One way to solve $L[y] = 0$ is to try a solution of the form $y = x^r$ (similar to our approach for solving constant coefficient equations, for which we tried $y = e^{rx}$). Let's try that...

Cauchy-Euler Equations

$$L[y] = a_n x^n \frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_{n-1} x^{n-1} \frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \cdots + a_1 x \frac{\mathrm{d}y}{\mathrm{d}x} + a_0 y = g(x) \qquad \text{(CE)}$$

where $a_n, a_{n-1}, \ldots, a_1, a_0 \in \mathbb{R}$ are constants with $a_n \neq 0$, and quantities in red are functions of the independent variable x.

On the interval $\mathcal{I} = (0, \infty)$, if $y = x^r$, then

$$\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = r(r-1)\cdots(r-n+1)x^{r-n}.$$

Substituting in $L[y] = 0$ and dividing by $x^r \neq 0$ gives

$$a_n r(r-1)(r-2)\cdots(r-n+1) + \cdots + a_2 r(r-1) + a_1 r + a_0 = 0\,.$$

This is the ***indicial equation*** for $L[y] = 0$ [Equation (CE)], analogous to the characteristic equation for constant coefficients.

Cauchy-Euler Equations

The most general *second order* Cauchy-Euler ODE is

$$L[y] = ax^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + bx\frac{\mathrm{d}y}{\mathrm{d}x} + cy = g(x) \qquad \text{(CE2)}$$

where $a, b, c \in \mathbb{R}$ are constants with $a \neq 0$, and quantities in red are functions of the independent variable x.

On the interval $\mathcal{I} = (0, \infty)$, $y = x^r$ is a solution of $L[y] = 0$ $\iff$

$$ar(r-1) + br + c = 0 \iff ar^2 + (b-a)r + c = 0$$

This is the ***indicial equation*** for the second order Cauchy-Euler equation [Equation (CE2)].

Just like the constant coefficient 2nd order ODE, there are three cases to consider: (i) distinct real roots, (ii) repeated root, and (iii) complex conjugate roots.

Cauchy-Euler Equations

Homogeneous second order Cauchy-Euler ODE:

$$L[y] = ax^2\frac{d^2y}{dx^2} + bx\frac{dy}{dx} + cy = 0\,, \qquad a, b, c \in \mathbb{R}$$

Indicial equation: $ar^2 + (b-a)r + c = 0$

Case (i): distinct real roots, $r_1, r_2 \in \mathbb{R}$

$$y_c = c_1 x^{r_1} + c_2 x^{r_2}$$

Case (ii): repeated root, $r \in \mathbb{R}$

$$y_c = c_1 x^r + c_2 x^r \ln x$$

Case (iii): complex conjugate roots, $\alpha \pm i\beta$, $\alpha \in \mathbb{R}$, $\beta > 0$

$$y_c = x^\alpha [c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)]$$

Cauchy-Euler Equations

Homogeneous second order Cauchy-Euler ODE:

$$L[y] = ax^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + bx\frac{\mathrm{d}y}{\mathrm{d}x} + cy = 0\,, \qquad a, b, c \in \mathbb{R}$$

Rather than guessing $y = x^r$, we could derive the solutions to $L[y] = 0$ by changing variables, *i.e.*, by changing the independent variable from x to $t = \ln x$ (which implies $x = e^t$).

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}t}\frac{1}{x} = \frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}t}$$

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{x}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right) = \frac{1}{x}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}t}\right)$$

$$= \frac{1}{x}\left(-\frac{1}{x^2}\frac{\mathrm{d}x}{\mathrm{d}t}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{1}{x}\frac{\mathrm{d}^2y}{\mathrm{d}t^2}\right) = \frac{1}{x}\left(-\frac{1}{x^2}\frac{1}{\mathrm{d}t/\mathrm{d}x}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{1}{x}\frac{\mathrm{d}^2y}{\mathrm{d}t^2}\right)$$

$$= \frac{1}{x}\left(-\frac{1}{x^2}x\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{1}{x}\frac{\mathrm{d}^2y}{\mathrm{d}t^2}\right) = \frac{1}{x^2}\left(\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - \frac{\mathrm{d}y}{\mathrm{d}t}\right)$$

Cauchy-Euler Equations

Homogeneous second order Cauchy-Euler ODE:

$$L[y] = ax^2\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + bx\frac{\mathrm{d}y}{\mathrm{d}x} + cy = 0\,, \qquad a, b, c \in \mathbb{R}$$

Changing variables from x to $t = \ln x$ we get

$$L[y] = ax^2\frac{1}{x^2}\left(\frac{\mathrm{d}^2y}{\mathrm{d}t^2} - \frac{\mathrm{d}y}{\mathrm{d}t}\right) + bx\frac{1}{x}\frac{\mathrm{d}y}{\mathrm{d}t} + cy = 0\,, \qquad a, b, c \in \mathbb{R}$$

Simplifying, we have

$$L[y] = a\frac{\mathrm{d}^2y}{\mathrm{d}t^2} + (b-a)\frac{\mathrm{d}y}{\mathrm{d}t} + cy = 0\,, \qquad a, b, c \in \mathbb{R}$$

which has *constant coefficients*, and yields the characteristic equation

$$ar^2 + (b-a)r + c = 0$$

for solutions $y = e^{rt} = e^{r\ln x} = e^{\ln x^r} = x^r$.
The characteristic equation is precisely the indicial equation found using the power trial $y = x^r$.

Cauchy-Euler Equations

Example ($x^2y'' + xy' - y = 0$)

This is a Cauchy-Euler equation with $a = 1$, $b = 1$, $c = -1$.

Try solution $y = x^r$, so $y' = rx^{r-1}$, $y'' = r(r-1)x^{r-2}$ and we get the *indicial equation*

$$r(r-1) + r - 1 = 0 \implies r^2 - 1 = 0 \implies r = \pm 1$$

and hence the general solution to this homogeneous equation is

$$y = c_1 x + c_2 x^{-1} = c_1 x + \frac{c_2}{x}$$

Cauchy-Euler Equations

Example ($x^2y'' - xy' + y = 0$)

This is a Cauchy-Euler equation with $a = 1$, $b = -1$, $c = 1$.

Try solution $y = x^r$, so $y' = rx^{r-1}$, $y'' = r(r-1)x^{r-2}$ and we get the *indicial equation*

$$\begin{aligned} r(r-1) - r + 1 = 0 &\implies r^2 - 2r + 1 = 0 \\ &\implies (r-1)^2 = 0 \\ &\implies r = 1 \quad \text{(repeated root)} \end{aligned}$$

and hence the *general solution* to this homogeneous equation is

$$y = c_1 x + c_2 x \ln x$$

Cauchy-Euler Equations

Example ($x^2y'' - xy' + y = \ln x$)

The associated homogeneous equation is the same as in the previous example, so the complementary function is

$$y_c = c_1 x + c_2 x \ln x \,.$$

How can we solve the non-homogeneous equation?

- Variation of parameters?
- Undetermined coefficients?
- Magic?

Please do *Cauchy–Euler method selection* on childsmath ...

Cauchy-Euler Equations

Example $(x^2y'' - xy' + y = \ln x)$

Given $L[c_1x + c_2x\ln x] = 0$, how can we solve $L[y] = \ln x$?

Method 1: Variation of Parameters

$$W = \begin{vmatrix} x & x\ln x \\ 1 & (\ln x)+1 \end{vmatrix},\quad W_1 = \begin{vmatrix} 0 & x\ln x \\ \frac{\ln x}{x^2} & (\ln x)+1 \end{vmatrix},\quad W_2 = \begin{vmatrix} x & 0 \\ 1 & \frac{\ln x}{x^2} \end{vmatrix}$$

Note: We had to put the equation in *standard form* before applying variation of parameters.

$$\Longrightarrow\quad W = x,\qquad W_1 = -\frac{(\ln x)^2}{x},\qquad W_2 = \frac{\ln x}{x}$$

$$\Longrightarrow\quad u_1 = \int \frac{-(\ln x)^2/x}{x}dx = -\int \frac{(\ln x)^2}{x^2}dx = \frac{1}{x}(\ln^2 x + 2\ln x + 2)$$

$$u_2 = \int \frac{(\ln x)/x}{x}dx = \int \frac{\ln x}{x^2}\,dx = -\frac{1}{x}(1+\ln x)$$

$$\Longrightarrow\quad y_p = \frac{1}{x}(\ln^2 x + 2\ln x + 2)x + -\frac{1}{x}(1+\ln x)x\ln x = \ln x + 2$$

$$\Longrightarrow\quad y = y_c + y_p = c_1x + c_2x\ln x + \ln x + 2$$

Cauchy-Euler Equations

Example ($x^2y'' - xy' + y = \ln x$)

Given $L[c_1x + c_2x\ln x] = 0$, how can we solve $L[y] = \ln x$?

Method 2: sneaky undetermined coefficients

First, change variables from x to $t = \ln x$.
This trick converts the ODE to $\ddot{y} - 2\dot{y} + y = t$.

Characteristic equation: $r^2 - 2r + 1 = 0 \implies (r-1)^2 = 0$
$\implies y_c = c_1e^t + c_2te^t$.

Try $y_p = At + B \implies \dot{y}_p = A \implies \ddot{y}_p = 0$
$\implies L[y_p] = \ddot{y}_p - 2\dot{y}_p + y_p = (0) - 2(A) + (At + B) = At + B - 2A$
Set $At + B - 2A = t$ to determine the coefficients A, B.
$\implies A = 1,\ B - 2A = 0 \implies B = 2A = 2$
$\implies y_p = t + 2 \implies y = y_c + y_p = c_1e^t + c_2te^t + t + 2.$
$\implies y = c_1x + c_2(\ln x)x + \ln x + 2$
as found by variation of parameters (with more effort).

Practice problems

Homogeneous Cauchy–Euler equations [Trench §7.4]

- Distinct real roots: 1–2, 7, 10–11, 15–16.
- Repeated roots: 3–4, 9, 13, 17.
- Complex conjugate roots: 5–6, 12, 14, 18.

Nonhomogeneous Cauchy–Euler equations [Trench §5.7]

- Particular solutions: 7, 11, 12, 16, 24.
- Initial-value problem and graph: 34.